BOARD OF INTERMEDIATE EDUCATION, KARACHI

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

CHEMISTRY

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

Section 'A' (Multiple Choice Questions)

Maximum Marks: 17

Note: (Options in the original are separated by star marks, shown here as A–D in printed order.)

Note: Time: 3 Hours — Maximum Marks: 85

Q1. Choose the correct answer for each from the given options:

Q1. Number of atoms in 60gm carbon will be:

(A) 3.01×10^{23} (B) 3.01×10^{24}

(C) 6.02×10^{23} (D) 6.02×10^{24}

Q2. Line spectra is used for the identification of a/an:

(A) Proton (B) Electron

(C) Element (D) Molecule

Q3. According to Hund's rule, the electronic configuration of carbon (C = 6) is:

(A) $1s^2\ 2s^2\ 2px^1,\ 2py^1$ (B) $1s^2\ 2s^2\ 2px^1\ 2py^1\ 2pz^1$

(C) $1s^2\ 2s^2\ 2px^2$ (D) $1s^2\ 2s^1\ 2px^1\ 2py^2$

Q4. The bond order in N_2 molecule is:

(A) 0 (B) 1

(C) 2 (D) 3

Q5. This molecule has zero dipole moment:

(A) NH_3 (B) HCl

(C) H_2O (D) CCl_4

Q6. Under similar conditions CH_4 gas diffuses faster than SO_2 gas:

(A) 1.5 times
(B) 2 times
(C) 4 times
(D) 16 times

Q7. It is an example of crystalline solid:

(A) Glass
(B) Rubber
(C) Table Salt
(D) Plastic

Q8. This liquid has the highest surface tension:

(A) Water
(B) Mercury
(C) Ethanol
(D) Gasoline

Q9. When a catalyst is added to a system equilibrium Kc will be:

(A) decreased
(B) increased
(C) unchanged
(D) zero

Q10. In equilibrium system $PCl_5 \rightleftharpoons PCl_3 + Cl_2$, the relationship between Kp & Kc is:

(A) Kp < Kc
(B) Kp > Kc
(C) Kp = Kc
(D) Kp ≠ Kc

Q11. This oxide is amphoteric in nature:

(A) K_2O
(B) CO_2
(C) CaO
(D) Al_2O_3

Q12. In pure water, universal indicator acquires this colour:

(A) Red
(B) Green
(C) Blue
(D) Pink

Q13. Rate constant of a chemical reaction is affected by:

(A) concentration of reactant
(B) temperature
(C) concentration of product
(D) reaction time

Q14. The sum of mole fractions of all the components of a solution is equal to:

(A) 1
(B) 10
(C) 100
(D) zero

Q15. Size of particles in a colloidal solution is:

(A) less than 1 nm
(B) between 1 nm and 1000 nm
(C) above 1000 nm
(D) zero

Q16. In a thermochemical process, work is zero if this is kept constant:

(A) Temperature (B) Volume

(C) Pressure (D) Mass

Q17. The outer body of dry cell is made up of:

(A) Copper (B) Zinc

(C) Lead (D) Iron

Section 'B' (Short-Answer Questions)

Maximum Marks: 36

Note: Answer any NINE part questions. All questions carry equal marks.

Note: ATOMIC MASS: (H = 1 am, C = 12 amu, Al = 27 amu, O = 16 amu, Cl = 35.5 amu, S = 32 amu, Ag = 108 amu, N = 14 amu)

Q2.

(i) Define any four of the following: ☆ Molar Volume ☆ Evaporation ☆ Bond Energy ☆ Limiting Reactant ☆ Dipole Moment

(ii) $AlCl_3$ is produced by the following reactant at 500°C: $2Al + 3Cl_2 \rightarrow 2AlCl_3$. When 160gm of Aluminium is reacted with excess of chlorine gas, 650gm of $AlCl_3$ is obtained. Calculate: (a) Theoretical yield (b) % yield

(iii) Combustion of Ethene (C_2H_4) in air to form CO_2 & H_2O is given in the following equation: $C_2H_4 + 3O_2 \rightarrow 2CO_2 + 2H_2O$. If a mixture containing 2.8 gm C_2H_4 & 6.4 gm O_2 is allowed to ignite, identify the limiting reactant and determine the mass of CO_2 gas that will be formed.

(iv) State Pauli's Exclusion Principle. Write electronic configuration of the following: ☆ Cl^- (Z = 17) ☆ Ca^{+2} (Z = 20) ☆ Cu (Z = 29)

(v) Draw the shape of BF_3, NH_3, CH_4 and H_2O molecules on the basis of VSEPR theory.

(vi) Differentiate between any two of the following: ☆ Crystalline Solid and Amorphous Solid ☆ Isomorphism and Polymorphism ☆ Molarity and Molality ☆ σ bond and π bond

(vii) 40 dm^3 of hydrogen gas is collected over water at 23°C and 831 torr pressure. What would be the volume of dry gas obtained at STP (vapour pressure of water at 23°C = 21 torr

(viii) Give scientific reasons for the following: ☆ At Mount Everest, atmospheric pressure falls down to about one-third, pressure at sea-level. ☆ Water droplets quickly deposit at the outer side of a glass of cold water. ☆ Mercury in a glass tube does not wet the walls. ☆ While boiling an egg, we often mix a small amount of table salt in water.

(ix) State Graham's Law of diffusion. The ratio of rates of diffusion of two gases A and B is 1.5:1 respectively. If the relative molecular mass of gas A is 16, find the molecular mass of B gas.

(x) Define Solubility Product. Write the solubility product expression for the following: ☆ $PbCl_2$ ☆ $Fe(OH)_3$ ☆ Ag_2SO_4

(xi) Define pH and pOH of a solution and write its formula. Calculate pOH of 0.0025M aqueous solution of HCl at 25°C. $HCl(l) \rightarrow H^+(aq) + Cl^-(aq)$

(xii) Calculate standard enthalpy of formation $\Delta Hf°$ of ($C_3H_8O_3$) from the data given below. ☆ $3C + 4H_2 + 3/2\ O_2 \rightarrow C_3H_8O_3$, $\Delta Hf° = ?$ ☆ $C + O_2 \rightarrow CO_2$, $\Delta Hf° = -393.5KJ/mole$ ☆ $H_2 + ½O_2 \rightarrow H_2O$, $\Delta Hf° = -285.8KJ/mole$ ☆ $C_3H_8O_3 + 3½O_2 \rightarrow 3CO_2 + 4H_2O$, $\Delta Hf° = -1654.1KJ/mole$

(xiii) What is order of reaction? Decomposition of NO_2 into NO and O_2 is 2nd order reaction: $2NO_2 \rightarrow 2NO + O_2$. If the rate constant at certain temperature is $3.8 \times 10^{-4}\ dm^3mol^{-1}s^{-1}$ and initial concentration of NO_2 is 0.38M, calculate the initial rate of reaction.

(xiv) State Raoult's Law. Derive this law mathematically in three forms. How this law can be applied on binary solution of miscible liquids? OR What is the fourth state of matter? State about its occurrence, characteristics and uses.

Section 'C' (Detailed Answer Questions)

Maximum Marks: 32

Note: Attempt any Two questions from this section. All questions carry equal marks.

Q3.

(a) Write the four main postulates of Bohr's atomic theory. Derive the relation $r_n = a°n^2/Z$ for the radius of nth orbit of Hydrogen atom.

(b) What is Hybridization? Write its types. CH_4, NH_3 & H_2O all have same hybridization but their molecular structures are different. Explaining their structures give a reason for this difference.

Q4.

(a) State Hess's Law? What is meant by Born-Haber's Cycle? Determine the lattice energy for CsF_s writing all the involved thermochemical equations stepwise, using following data: (all values are in KJ/mole) ☆ $\Delta H°f$ of CsF = -553.5 ☆ ΔH(sub) of Cs = 76.5 ☆ ΔH_BA of F = -328.2 ☆ ΔH(sub) of Cs = 375.5 ☆ ΔH_diss of F_2 = 157

(b) Give the statement and mathematical derivation of First Law of Thermodynamics and derive $q_v = \Delta E$ and $q_p = \Delta H$

Q5.

(a) Define Oxidation and Reduction according to modern electronic concept. Balance the following equation by ion electron method: ☆ $Cr_2O_7^{-2} + Cl^- \rightarrow + Cr^{+3} + Cl_2$ (Acidic medium) ☆ $MnO_4^- + SO_3^{-2} \rightarrow Mn^{+2} + SO_4^{-2}$ (Basic medium)

(b) Define Cell Potential, Standard Electrode Potential. Explain the determination of standard electrode Zinc with diagram and potential of cell reactions.